



FINAL EXAMINATION SEMESTER III, SESSION 2021/2022

COURSE CODE : FSPP 0034
COURSE : PHYSICS II
PROGRAMME : FOUNDATION UTM
DURATION : 3 HOURS
DATE : 26th of APRIL 2022

INSTRUCTION TO CANDIDATES:

1. ANSWER ALL QUESTIONS.

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This examination question consists of (8) printed pages only including this page.

PART A

Question 1 (20 MARKS)

- a) Figure 1 depicts a section of a cylindrical conducting wire containing some free electrons. The wire is connected to an emf source, and the electric field, E is oriented as shown in the figure.

At random temperature, the electrons in a conductor are moving randomly in all direction. When the conductor is connected to an electric source, the electrons are attracted to the positive terminal of the source and they are moving with drift velocity

- Sketch a diagram to show the motions of one free electron, the direction of drift velocity and the current.
- Briefly explain the drift velocity of an electron of the conductor.
- Calculate the drift velocity of an electron in the conducting wire with a cross-sectional area of 1.20 mm^2 and carrying a current capacity of 2.8 A with 10^{29} free electrons per metre cubic of the wire. $1.46 \times 10^{-4} \text{ m/s}$

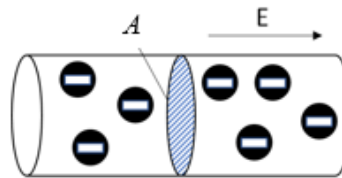


Figure 1

(6 marks)

- b) A 12 V battery is connected across a 7Ω resistor. The terminal voltage is 11.7 V . Calculate:

- The current through the resistor. 1.67 A
- Power dissipated by the resistor. 19.52 W
- Internal resistance of the battery. 0.18 ohm

(6 marks)

- c) Figure 2 shows a circuit consisting of two batteries and three resistors. Determine:

- The value of I_1 , I_2 and I_3 . $0.25 \text{ A}, 0.42 \text{ A}, 0.67 \text{ A}$
- The potential difference across the 18Ω resistor. 6.53 V

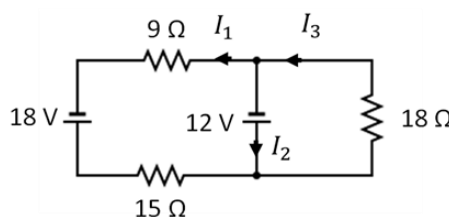
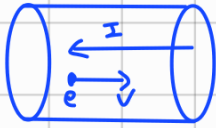


Figure 2

(8 marks)

1) (a) (i)



$$\text{ciii) drift velocity, } v_d = \frac{I}{neA} = \frac{2.8}{1 \times 10^{29} (-1.6 \times 10^{-19}) (1.2 \times 10^{-6})} \\ = 1.46 \times 10^{-4} \text{ m/s}$$

$$(b) \quad \mathcal{E} = 12 \text{ V}, R = 7 \Omega, V_{\text{terminal}} = 11.7 \text{ V}$$

$$\text{ci) } V = IR$$

$$11.7 = I(7)$$

$$I = 1.67 \text{ A}$$

$$\text{cii) } P = IV$$

$$= 1.67(11.7)$$

$$= 19.54 \text{ W}$$

$$\text{ciii) } \mathcal{E} = V + Ir$$

$$12 = 11.7 + 1.67r$$

$$r = 0.18 \Omega$$

$$(c) \text{ ci) } I_1 + I_2 = I_3 \quad - \textcircled{1}$$

$$18 - 12 = 9I_1 + 15I_1$$

$$6 = 24I_1, \quad 2 = 3I_3, \quad I_3 = \frac{2}{3} \text{ A} \quad \#$$

$$6 = 24I_1$$

$$4I_1 = 1, \quad I_1 = \frac{1}{4} \text{ A}$$

$$\frac{1}{4} + I_2 = \frac{2}{3}$$

$$I_2 = \frac{2}{3} - \frac{1}{4}$$

$$= \frac{5}{12} \text{ A} \quad \#$$

Question 2 (20 MARKS)

i. No.
Sound is a longitudinal mechanical wave. It requires the presence of a medium; electromagnetic waves do not require a medium.

- a) i. Is sound an electromagnetic wave? If not, what kind of wave is it?
 ii. Are the wavelengths of radio and television signals longer or shorter than those detectable by the human eye?
 The wavelengths of radio signals are longer than those of visible light.
 The wavelengths of television signals are longer than those of visible light. (4 marks)

- b) An EM wave has frequency 9.76×10^{14} Hz. What is its wavelength, and how would we classify it? 3.07×10^{-7} m, UV

(2 marks)

- c) A solenoid has 400 turns, cross sectional area of $3.5 \times 10^{-3} \text{ m}^2$ and length 20.0 cm. A current of 4.3 A flows through the solenoid. Calculate:

- i. The inductance of the solenoid. 3.52×10^{-3} H
 ii. The energy stored in the solenoid. 0.033 J
 iii. The induced emf in the solenoid if the current is reduced uniformly in 70 ms. 0.22 V

(6 marks)

- d) Table 1 shows the data for two coils X and Y. It is given that the mutual inductance between the coils is 0.007 H. At a particular instant, the current in the coil X is 0.006 A and the rate change of current is 2 As^{-2} .

$$M = \frac{N_2 \Phi_1}{I_1} = \frac{300(\Phi_2)}{0.006} = 0.007$$

Table 1

	Coil X	Coil Y
Number of turns	250	300
Self inductance	0.017 H	-

Calculate:

- i. The magnetic flux linkage in coil X. 4.08×10^{-7} Wb
 ii. The back emf in coil X. 0.034 V
 iii. The magnetic flux linkage in coil Y. 1.4×10^{-7} Wb
 iv. The mutually induced emf in coil Y. 0.014 V

(8 marks)

2) (b) $f = 9.76 \times 10^{14} \text{ Hz}$, $c = f\lambda$
 $3 \times 10^8 = 9.76 \times 10^{14} \lambda$
 $\lambda = 3.07 \times 10^{-7} \text{ m}$, ultraviolet

(c) (i) $L = \frac{\mu_0 N^2 A}{\lambda} = \frac{4\pi \times 10^{-7} (400^2) (35 \times 10^{-3})}{0.2}$
 $= 3.52 \times 10^{-3} \text{ H}$

(ii) $U = \frac{1}{2} \left(\frac{\mu_0 N^2 A I^2}{\lambda} \right) = \frac{1}{2} (3.52 \times 10^{-3} (4.3^2))$
 $= 0.0325 \text{ J}$

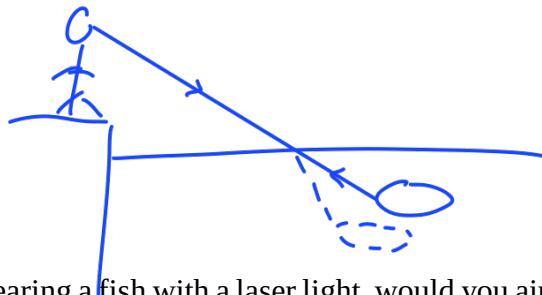
(iii) $\mathcal{E} = -L \frac{dI}{dt}$
 $= -3.52 \times 10^{-3} \left(\frac{0 - 4.2}{70 \times 10^{-3}} \right)$
 $= 0.216 \text{ V}$

(d) (i) $L = \frac{N^2 \Phi}{I}$
 $0.017 = \frac{250 \Phi}{0.006}$
 $\Phi = 4.08 \times 10^{-7} \text{ Tm}^{-2}$

(ii) $\mathcal{E} = -L \frac{dI}{dt}$
 $= -0.017(2)$
 $= -0.034 \text{ V}$

(iii) $M = \frac{N_2 \Phi_2}{I_1}$
 $0.007 = \frac{300(\Phi_2)}{0.006}$
 $\Phi_2 = 1.4 \times 10^{-7}$

(iv) $\mathcal{E} = -M \frac{dI}{dt}$
 $= -0.007(2)$
 $= -0.014 \text{ V}$

**Question 3 (20 MARKS)**

- a) i. If you were spearing a fish with a laser light, would you aim directly at the observed fish, slightly above or slightly below in order to make a direct hit. Explain your answer.
 You should aim directly at the observed fish. The laser light will refract as it enters the water, following the exact reverse path as the light coming from the fish to your eye.
- ii. An object is placed along the principal axis of a spherical mirror. The magnification of the object is -3. Is the image real or virtual, inverted or upright? Is the mirror concave or convex?

$m = -\frac{d_i}{d_o} = -3$
 d_i must be (+) → real image
 The image is real and inverted, because the magnification is negative. The mirror is concave, because convex mirrors can only form virtual image (4 marks)

- b) i. A convex mirror forms a virtual image half the size of the object. Assuming the distance between the image and the object is 20.0 cm. Determine the radius of curvature of the mirror. 26.7 cm
- ii. A beam of light travelling in air is incident perpendicularly on a prism with a deviation angle of 20° as shown in Figure 3. Determine the index of refraction of the prism. 1.28

(Given: $c = 3 \times 10^8 \text{ ms}^{-1}$, $n_{\text{air}} = 1.00$)

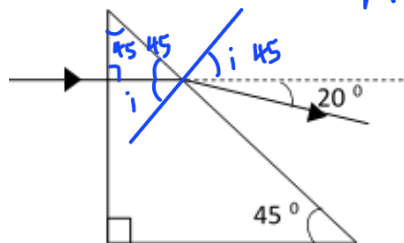


Figure 3

$$m = \frac{1}{2} = -\frac{d_i}{d_o} \quad \frac{d_o - d_i}{d_o} = \frac{1}{2}$$

$$40 - 2d_i = 20$$

$$3d_i = 40$$

$$i = 45$$

$$r = 65$$

(7 marks)

- c) A diverging lens with a focal length of -15.0 cm is placed 12.0 cm to the right of the converging lens with a focal length of 18.0 cm. The final image that formed the virtual image is located 37.5 cm to the left of the diverging lens. Determine the position of the object that is placed to the left of the converging lens. 35 cm

(5 marks)

3) (b) (i) $m = \frac{1}{2} = -\frac{d_i}{d_o} \quad \text{--- (1)}$ $\frac{1}{f} = \frac{3}{40} + \left(-\frac{3}{20}\right) = -\frac{40}{3}$ (convex mirror)

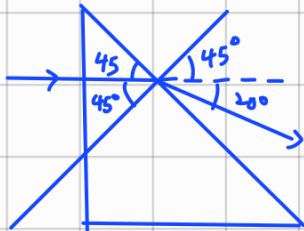
$d_o + d_i = 20 \quad \text{--- (2)}$ $C = 2f = 2\left(\frac{40}{3}\right)$
 $-\frac{d_i}{d_o} = \frac{1}{2}$ $= \frac{80}{3}$
 $-2d_i = d_o$ $= 26.6 \text{ cm}$

$-3d_i = 20$

$d_i = -\frac{20}{3}$

$d_o = \frac{40}{3}$

(ii)

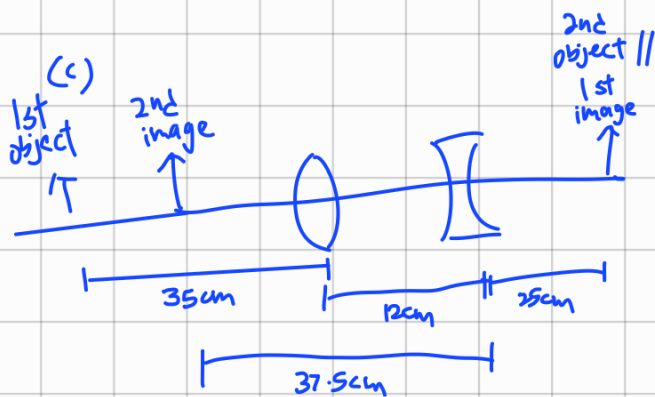


from glass to air

$\frac{n_2}{n_1} = \frac{\sin i}{\sin r}$

$\frac{1.00}{n_1} = \frac{\sin(45^\circ)}{\sin(65^\circ)}$

$n_{\text{glass}} = 1.282$



Concave lens

$\frac{1}{-15} = \frac{1}{d_o} + \frac{1}{-37.5}$

$d_o = -25 \text{ cm}$ (back of the lens)

Convex lens

$d_i = +37 \text{ cm}$

$\frac{1}{18} = \frac{1}{d_o} + \frac{1}{37}$

$d_o = 35 \text{ cm}$

$m = m_1 \times m_2$

$= -\frac{d_i}{d_o} \times -\frac{d_i}{d_o}$

$= 1.586 \rightarrow \text{upright, magnified, virtual}$

(d) (i) $\frac{1}{f} = (n-1)\left(\frac{1}{r_1} + \frac{1}{r_2}\right)$

$\frac{1}{230 \times 10^{-3}} = (1.54-1)\left(\frac{1}{r_1} + \frac{1}{\infty}\right)$

$\frac{1}{230 \times 10^{-3}} = \frac{0.54}{r_1}$

$r_1 = 0.54(230 \times 10^{-3})$

$= 0.1242 \text{ m}$

(ii) $\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$

$\frac{1}{0.23} = \frac{1}{2} + \frac{1}{d_i}$

$d_i = 0.26 \text{ m}$



- d) A lens maker plans to construct a plano convex lens with a focal length of 230.0 mm. Given the refractive index of the lens is 1.54. Determine:

- i. The radius of curvature for the curved surface of the lens. 0.124 m
- ii. Position of the image if it focuses an object 2.0 m away. 0.26 m

(4 marks)

APPENDIX 1

List of constant

$$p = +1.60 \times 10^{-19} \text{ C} \quad -$$

$$e = -1.60 \times 10^{-19} \text{ C}$$

$$\mu_o = 4\pi \times 10^{-7} \text{ m A}^{-1}$$

$$\varepsilon_o = 8.85 \times 10^{-12} \text{ C}^2 \text{N}^{-1} \text{m}^{-2}$$

$$g = 9.80 \text{ ms}^{-2}$$

$$c = 3 \times 10^8 \text{ ms}^{-2}$$

$$m_e = 9.11 \times 10^{-31} \text{ kg}$$

$$k = 9.0 \times 10^9 \text{ Nm}^2 \text{ C}^2$$

$$h = 6.63 \times 10^{-34} \text{ J s}$$

$$N_A = 6.022 \times 10^{23}$$

$$1 \text{ u} = 1.6605 \times 10^{-27} \text{ kg} = 931.5 \text{ MeV}/c^2$$

$$1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$$

Refractive index of air, $n = 1.00$

Nuclide	Mass, m (u)
${}^2_1\text{H}$	2.014102
${}^3_1\text{H}$	3.016049
${}^1_1\text{H}$	1.007825

APPENDIX 2

List of Formulae

$$B_o = \frac{E_o}{c}$$

$$a \sin \theta = n \lambda$$

$$E = KE + W_o$$

$$U = \frac{1}{2} L I^2$$

$$\bar{P} = I_{rms} V_{rms}$$

$$I_{in} = I_{out}$$

$$R_T = R_o [1 + \alpha (T - T_o)]$$

$$\tau = RC$$

$$\tau = \frac{L}{R}$$

$$V_{rms} = I_{rms} R$$

$$V_c = V_o e^{-t/\tau}$$

$$I = n A v e$$

$$M = \frac{\mu_o N_1 N_2 A}{l}$$

$$L = \frac{\mu_o N^2 A}{l}$$

$$L = \frac{N \Phi}{I}$$

$$\mathcal{E} = \left| -L \frac{dI}{dt} \right|$$

$$\lambda = \frac{c}{f}$$

$$k = \frac{2\pi}{\lambda}$$

$$\omega = 2\pi f$$

$$m = -\frac{d_i}{d_o} = \frac{h_i}{h_o}$$

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

$$\frac{\sin \theta_i}{\sin \theta_r} = \frac{n_2}{n_1}$$

$$\frac{1}{f} = (n-1) \left[\frac{1}{r_1} + \frac{1}{r_2} \right]$$

$$W_o = h f_o$$

$$N = N_o e^{-\lambda t}$$

$$p = mv$$

$$KE = eV$$

$$T_{1/2} = \frac{\ln 2}{\lambda}$$

$$Q = (\Delta m) c^2$$

$$d \sin \theta = m \lambda$$

$$m = 0, 1, 2, 3, \dots$$

$$d \sin \theta = (m + \frac{1}{2}) \lambda$$

$$m = 0, 1, 2, 3, \dots$$

$$Q = Q_o e^{-t/\tau}$$

$$I = I_o e^{-t/\tau}$$

$$\lambda = \frac{\ln 2}{T_{1/2}}$$

$$\left| \frac{dN}{dt} \right|_o = \lambda N_o$$

$$\left| \frac{dN}{dt} \right| = \left| \frac{dN}{dt} \right|_o e^{-\lambda t}$$